Arno Verduyn

Curriculum Vitae

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Summary

Engineer and researcher with a PhD in mechanical engineering and a background in electrical engineering. Specialized in the analysis and representation of rigid-body motion, with a strong focus on mathematical modeling, numerical robustness, and dealing with edge cases and imperfect data. Experienced in solving theoretically complex problems through careful reasoning and clear communication within multidisciplinary engineering environments.

Professional Experience

2025–Present **Postdoctoral Researcher — Robotics Research Group, Department of Mechanical Engineering**, *KU Leuven*, Leuven, Belgium.
- Continued research and formalization of invariant representations of rigid-body motion, resulting in technical academic publications.
- Co-developed research project proposals; authored conference and workshop papers; presented research at international conferences.

2020–2025 **Doctoral Researcher — Robotics Research Group, Department of Mechanical Engineering**, *KU Leuven*, Leuven, Belgium.
- Developed novel invariant representations for rigid body motion for robustly comparing motion trajectory data under changes in reference frame and measurement noise.
- Interpreted experimental and simulated data, with particular attention to edge cases, representation singularities, and model limitations.
- Contributed to the ERC Advanced Grant project "Robotgenskill", with a focus on generalization and robustness in robot skill learning.
- Participated in an applied Flemish research project on trajectory generation for robotic spray painting in an industrial context.
- Worked within a multidisciplinary research team, providing analytical support and methodological input.
- (Co-)mentored master thesis students, guiding reasoning, methodology, and interpretation of results.
- (Co-)authored peer-reviewed journal and conference papers.
- Presented papers at international conferences.
- Assisted with undergraduate teaching and evaluation.

Degrees

2025 **PhD in Mechanical Engineering**, *KU Leuven*, Leuven, Belgium.
Thesis: *Invariant Trajectory Similarity Measurement: Resolving Singularity Issues for Robust Rigid-Body Motion Recognition*

2020 **Master of Science in Mechanical Engineering**, *KU Leuven*, Leuven, Belgium.
Specialization in Mechatronics and Robotics. Graduated *Cum Laude*. (Master thesis: *Summa Cum Laude*)

2018 **Bachelor of Science in Electrical Engineering**, *KU Leuven*, Leuven, Belgium.
Graduated *Cum Laude*.

Engineering and Analytical Competencies

- Mathematical modeling and interpretation
- Robustness analysis and handling of edge cases
- Linear algebra, differential geometry, numerical methods, and optimization
- Technical documentation and clear explanation of methods and assumptions

Programming and Tools

- **MATLAB** and **Python**: numerical computing, modeling, simulation development, algorithm prototyping
- **C++**: basic algorithm implementation
- **Git**: version control

Languages

Dutch	Native	*Mother tongue*
English	Fluent	*Used in academic and professional settings*
French	Conversational	*Can understand and communicate in routine situations.*

Award(s)

Best Poster Award at the 2023 Flanders Make Scientific conference on machines, vehicles, and production technology.

References

Available upon request.